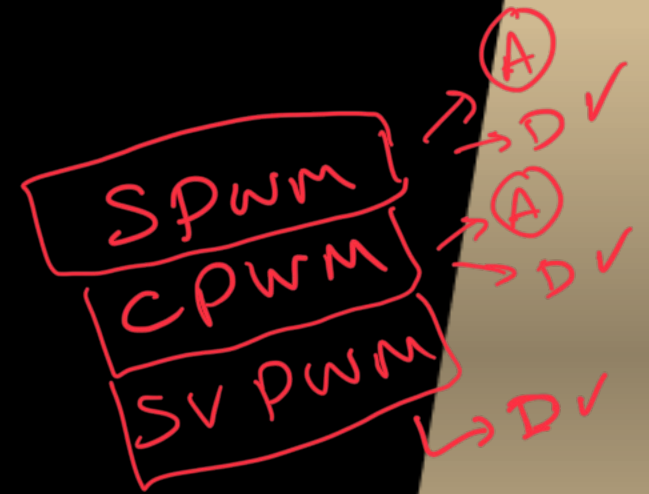


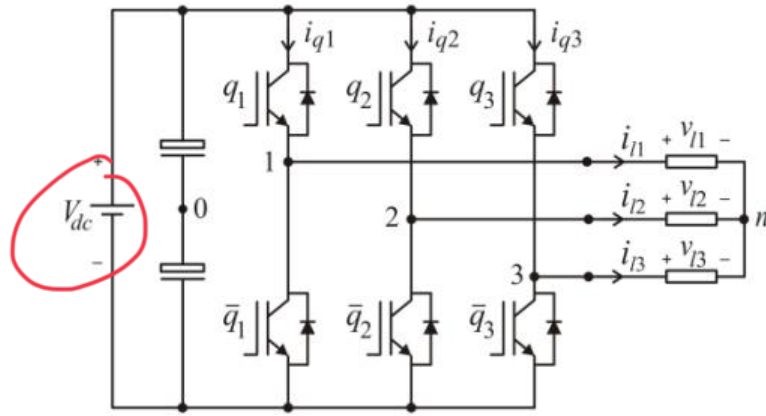
ECE595: Power Converters for AC Systems

Lecture 21 (PWM cont.)

Prof. Euzeli Dos Santos



3. Three-leg Converter and Three-phase Load



$$v_{10} = (2q_1 - 1) \frac{V_{dc}}{2} \quad \checkmark$$

$$v_{20} = (2q_2 - 1) \frac{V_{dc}}{2} \quad \checkmark$$

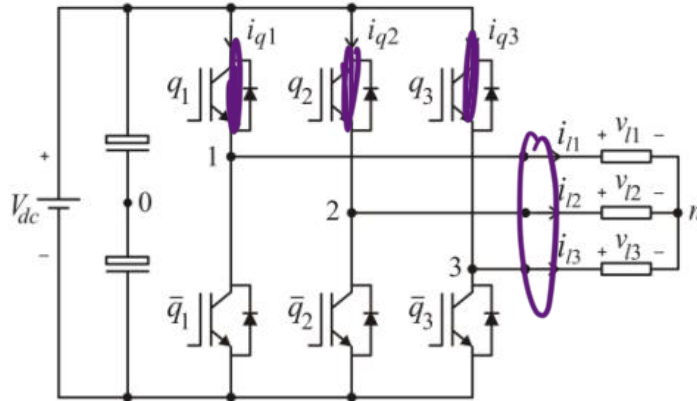
$$v_{30} = (2q_3 - 1) \frac{V_{dc}}{2} \quad \checkmark$$

3.2. PWM Implementation

$$\begin{aligned} \rightarrow v_{l1}^* &= v_{10}^* - v_{n0}^* \\ \rightarrow v_{l2}^* &= v_{20}^* - v_{n0}^* \\ \rightarrow v_{l3}^* &= v_{30}^* - v_{n0}^* \end{aligned}$$

$$\left[\begin{aligned} v_{l1}^* &= V_l^* \cos(\omega t) \\ v_{l2}^* &= V_l^* \cos(\omega t + 120^\circ) \\ v_{l3}^* &= V_l^* \cos(\omega t - 120^\circ) \end{aligned} \right.$$

3. Three-leg Converter and Three-phase Load



$$\begin{aligned} v_{10}^* &= v_{l1}^* + v_h^* \\ v_{20}^* &= v_{l2}^* + v_h^* \\ v_{30}^* &= v_{l3}^* + v_h^* \end{aligned}$$

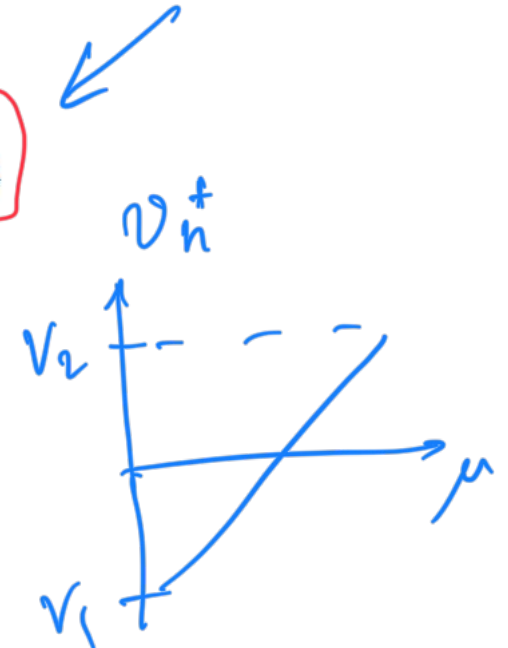
$$\begin{aligned} v_{10}^* &= v_{l1}^* + v_h^* \\ v_{20}^* &= 0 + v_h^* \end{aligned}$$

$$v_h^* = V_{dc} \left(\mu - \frac{1}{2} \right) + (\mu - 1)V_{max}^* - \mu V_{min}^*$$

$$V_{max} = \text{MAX}\{v_{l1}^*, v_{l2}^*, v_{l3}^*\}$$

$$V_{min} = \text{MIN}\{v_{l1}^*, v_{l2}^*, v_{l3}^*\}$$

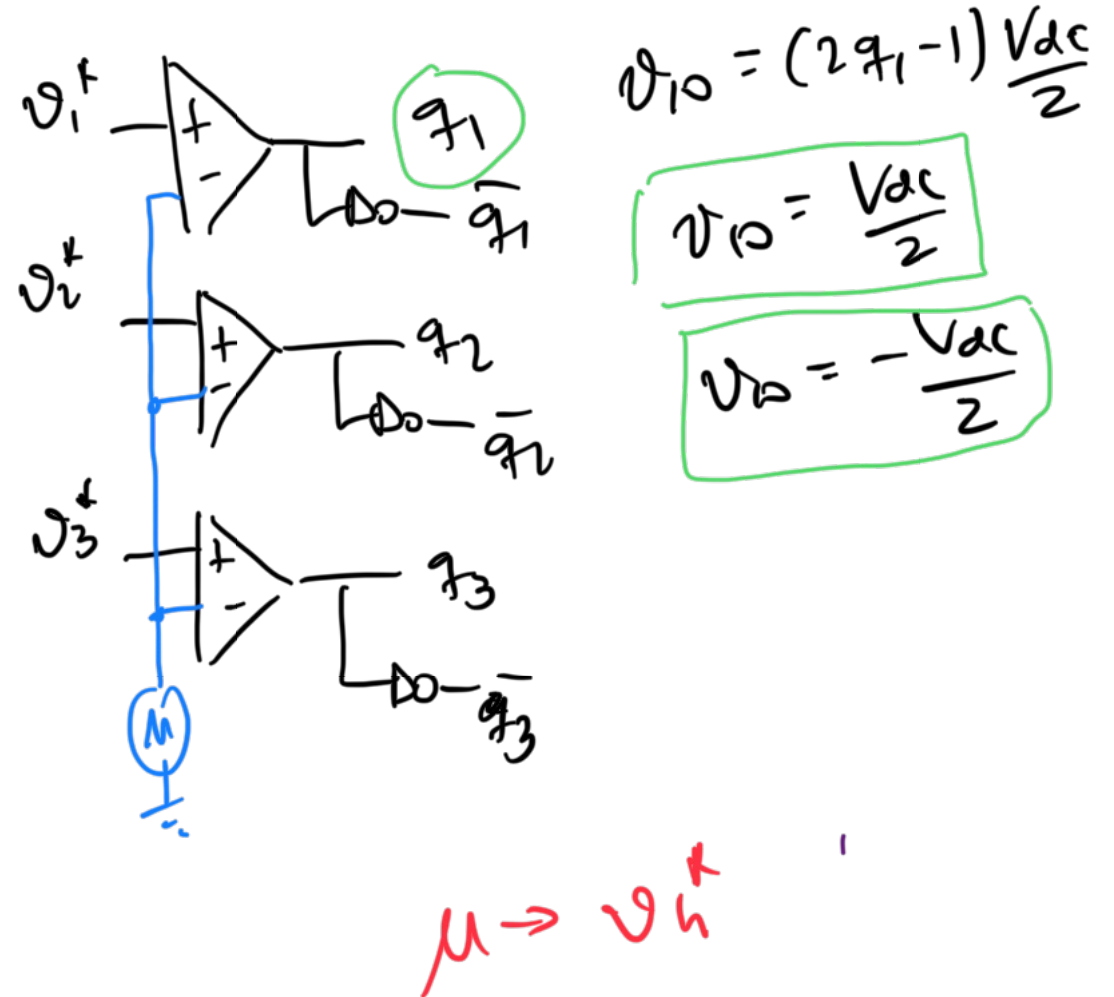
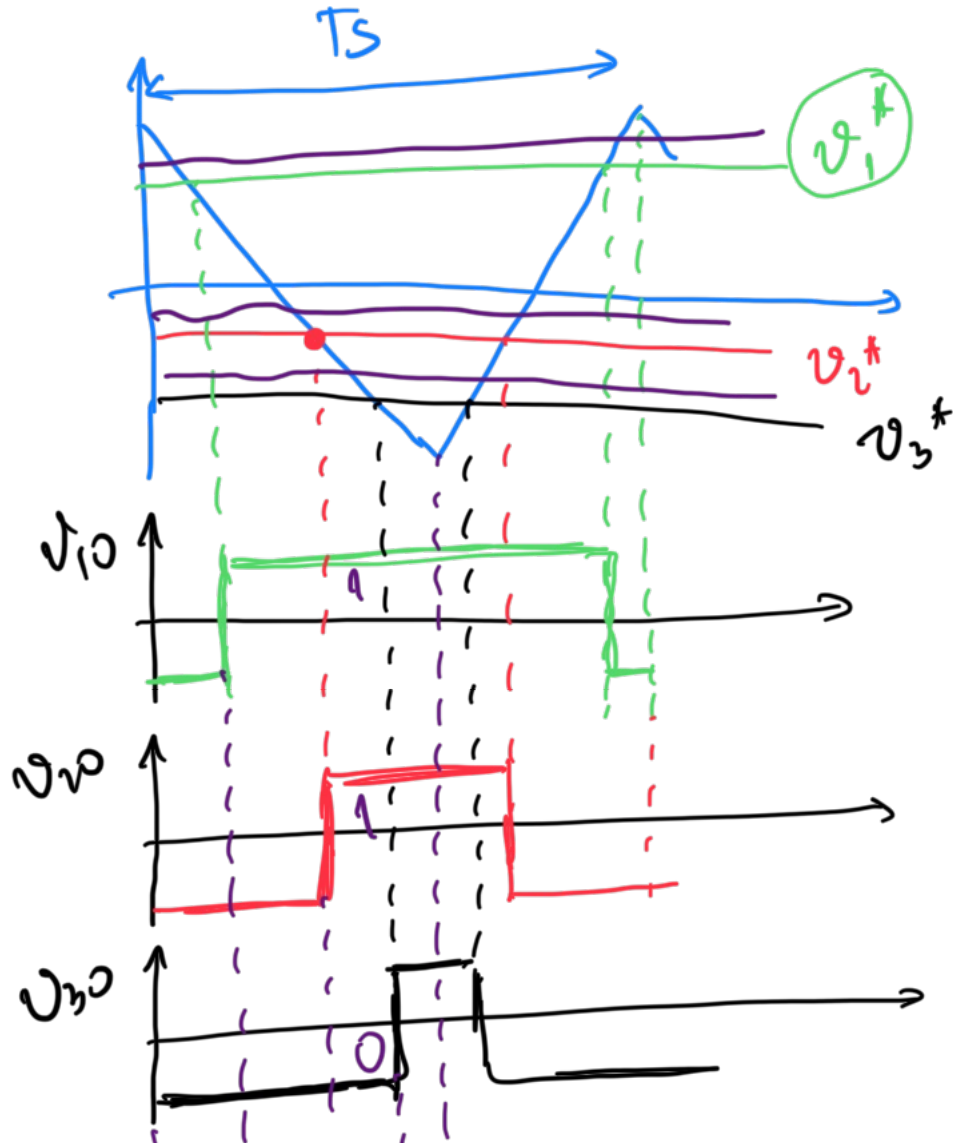
μ is the distribution factor and determines how the free-wheeling time interval can be distributed throughout the switching period.



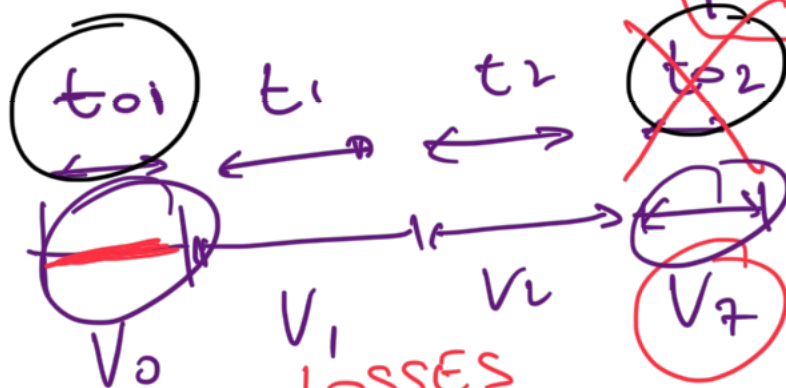
The Concept of Distribution Ratio $0 \leq \mu \leq 1$

When a zero-sequence component is added to each sinusoidal reference, new references are generated.

Note that the time of active vectors remain unchanged



	q_1	q_2	q_3	
①	0	0	0	Zero vector (V_0)
②	1	0	0	Active vector (V_1)
③	1	1	0	Active vector (V_2)
④	1	1	1	Zero vector (V_7)



$$\mu = \frac{t_{02}}{t_{01} + t_{02}}$$

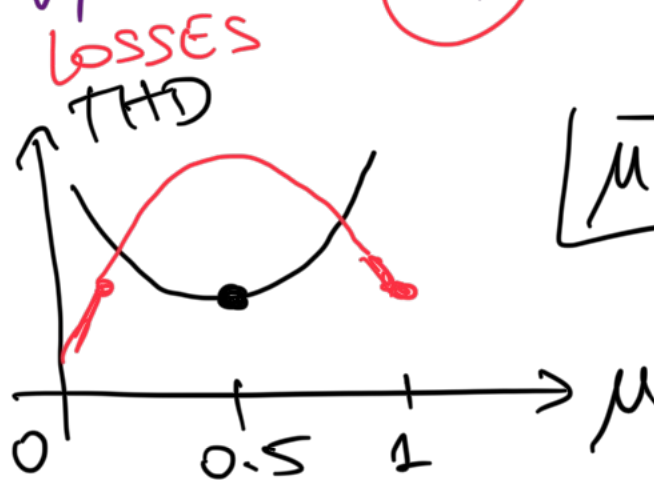
$\mu = 0$
 $t_{02} = 0$

$$\mu = 0.5$$

$$t_{01} = t_{02}$$

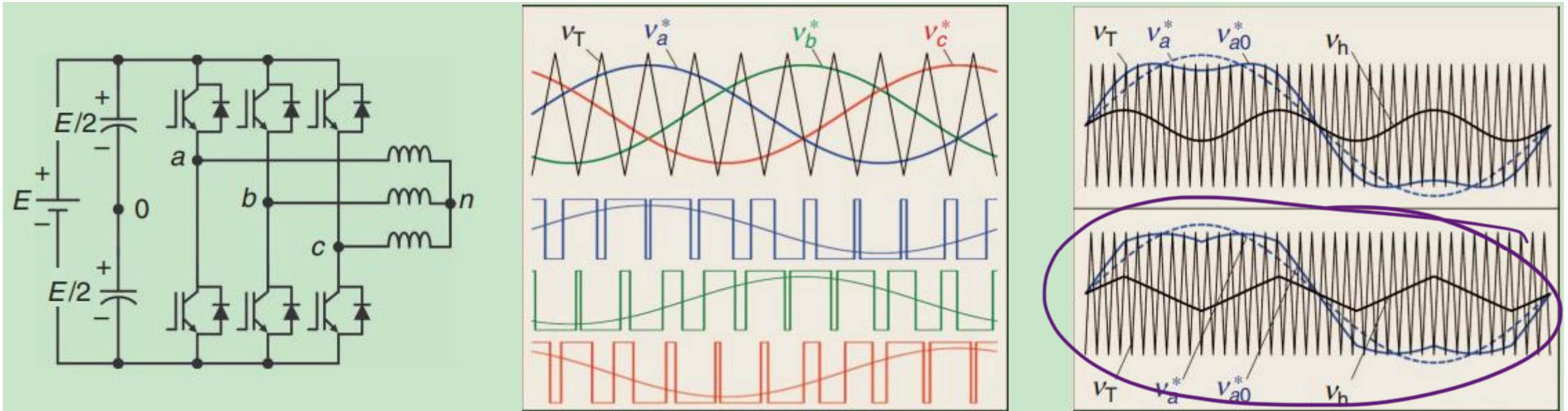
$$\mu = 1$$

$$t_{01} = 0$$

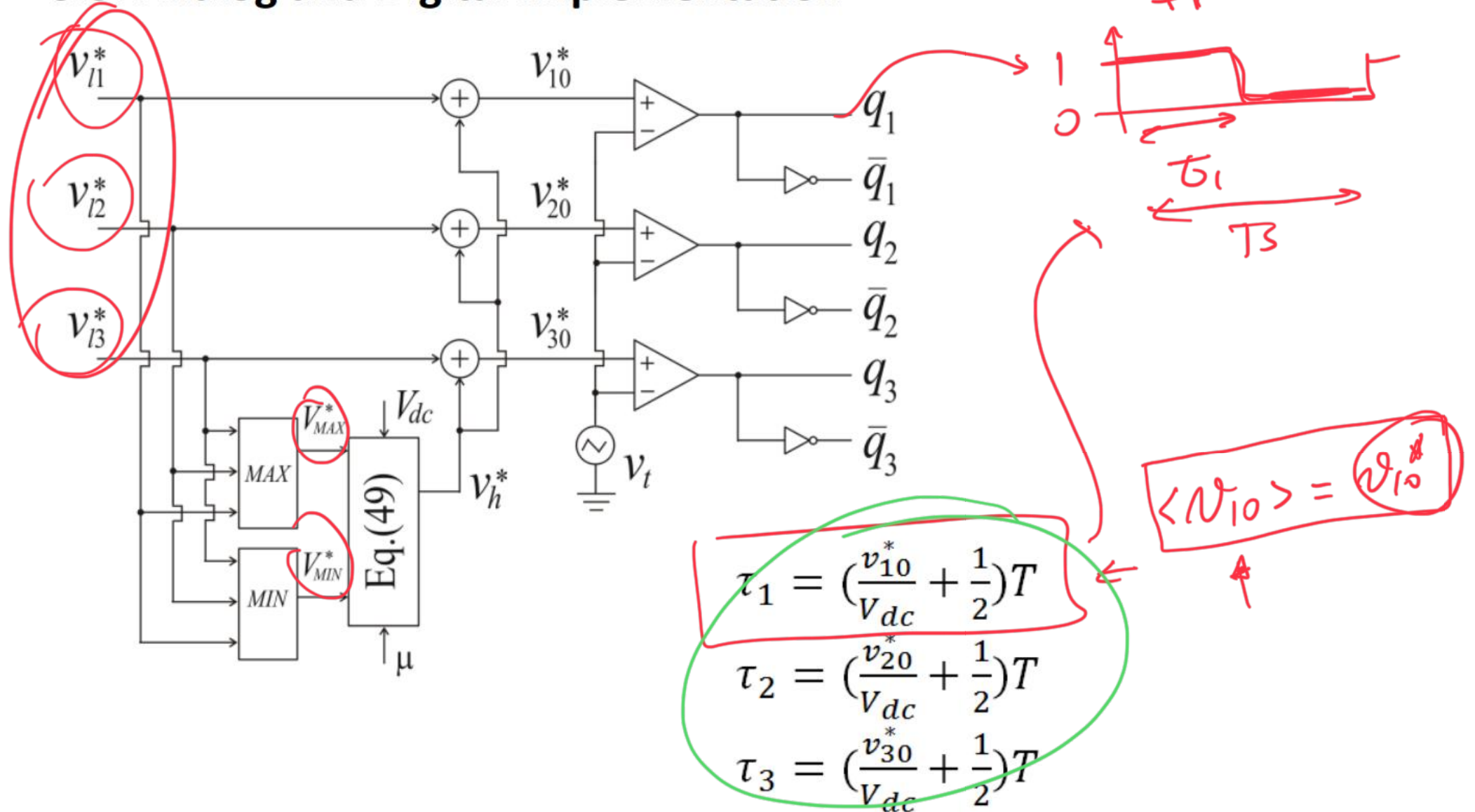


$$v_h^* = V_{dc} \left(\mu - \frac{1}{2} \right) + (\mu - 1)V_{max}^* - \mu V_{min}^*$$

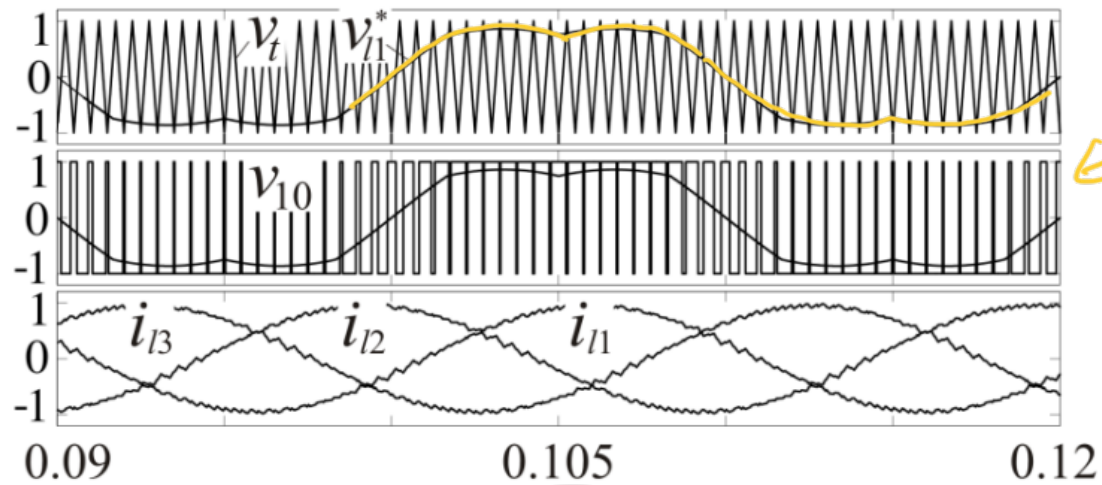
3dr



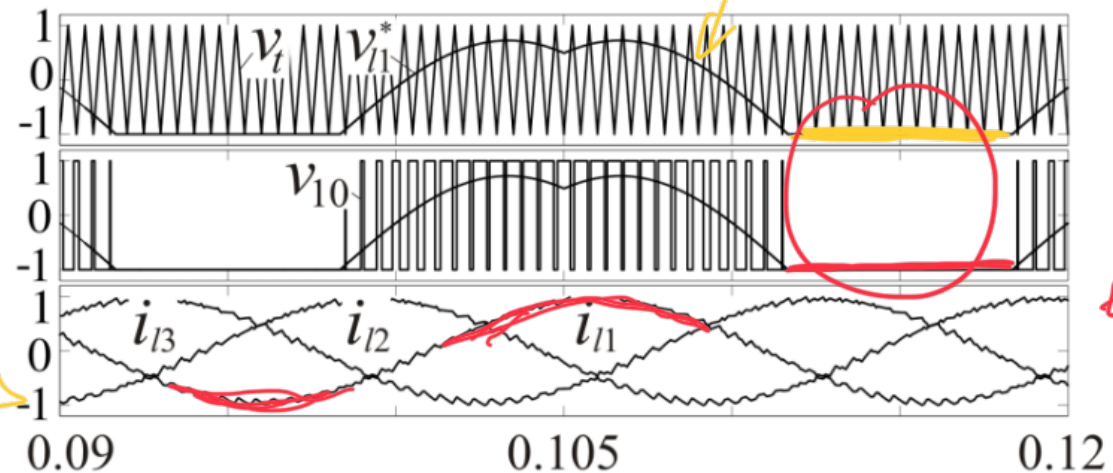
3.3. Analog and Digital Implementation



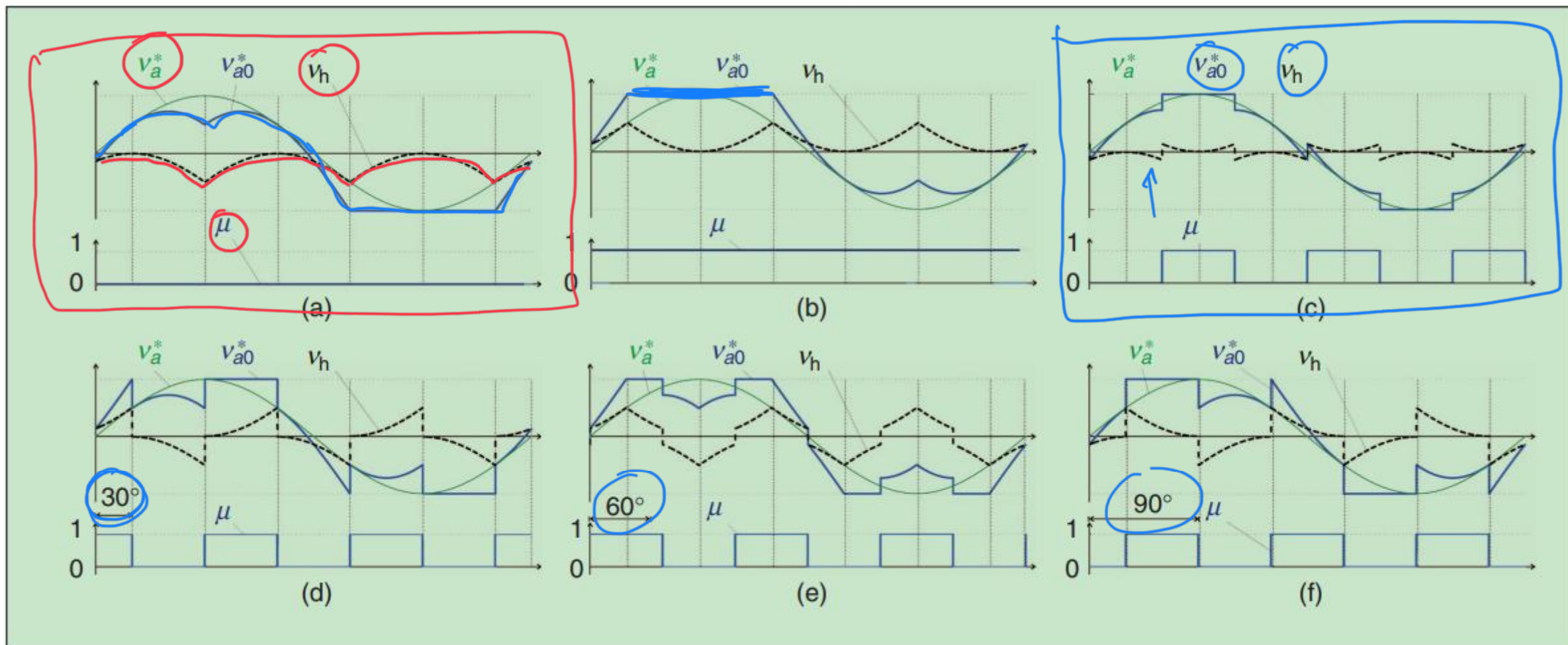
3.4. Influence of μ for PWM implementation



Waveforms with $\mu = 0.5$.



Waveforms with $\mu = 1.0$.



```
while t<=tmax,
    t = t + h; % simulation time
```

```
tetal=tetal+h*w1;
if tetal >= 2*pi,
    tetal=tetal-2*pi; %resets theta to avoid very large numbers
end
```

```
vas_ref = 1*24*cos(tetal); %reference voltage - phase a
vbs_ref = 1*24*cos(tetal-2*pi/3); %reference voltage - phase b
vcs_ref = 1*24*cos(tetal-4*pi/3); %reference voltage - phase c
```

```
%begin of the pwm loop simulation (interruption)
if t>tpwm,
```

```
    tpwm=tpwm+Ts;
```

```
    cont=0;
```

```
    tal1=(vas_ref/Vdc+0.5)*Ts;
```

```
    tal2=(vbs_ref/Vdc+0.5)*Ts;
```

```
    tal3=(vcs_ref/Vdc+0.5)*Ts;
```

```
end
```

```
%end of the pwm loop simulation
```

```
cont=cont+h;
```

```
%begin simulation of the converter
```

```
if (cont<tal1)
```

```
    q1=1;
```

```
else
```

```
    q1=0;
```

```
end
```

```
if (cont<tal2)
```

```
    q2=1;
```

```
else
```

```
    q2=0;
```

```
end
```

```
if (cont<tal3)
```

```
    q3=1;
```

```
else
```

```
    q3=0;
```

```
end
```

```
%Definition of the pole voltages
```

```
v10 = (2*q1-1)*(Vdc/2);
```

```
v20 = (2*q2-1)*(Vdc/2);
```

```
v30 = (2*q3-1)*(Vdc/2);
```

```
vn0 = (1/3)*(v10+v20+v30);
```

```
%Phase voltages applied to the stator of the machine
```

```
vas = v10 - vn0;
```

```
vbs = v20 - vn0;
```

```
vcs = v30 - vn0;
```

```
%Euler integration method to obtain currents
```

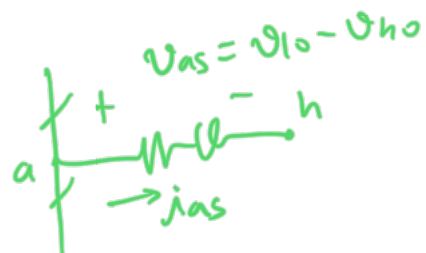
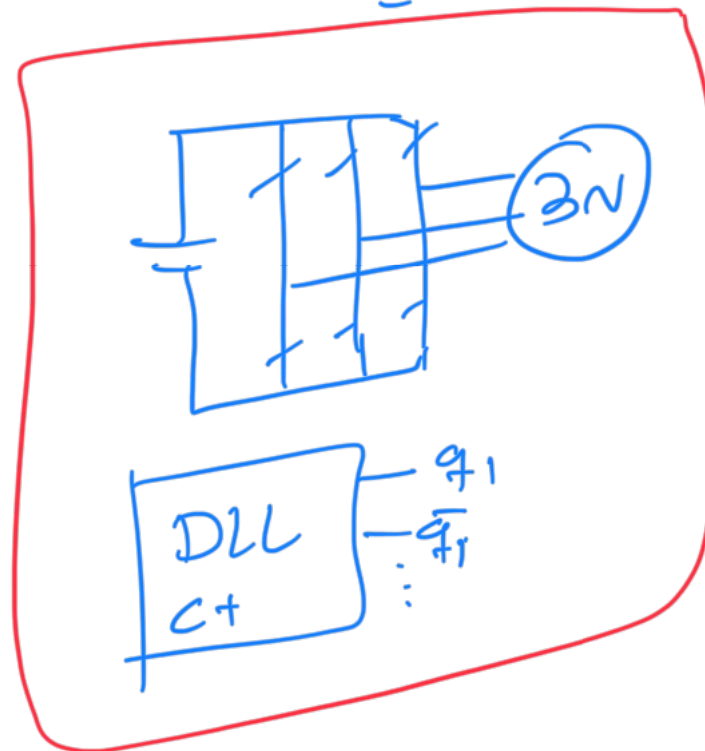
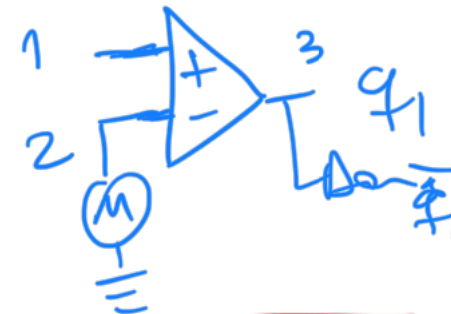
```
ias = ias + (vas - rs*ias)*h/ls;
```

```
ibs = ibs + (vbs - rs*ibs)*h/ls;
```

```
ics = ics + (vcs - rs*ics)*h/ls;
```

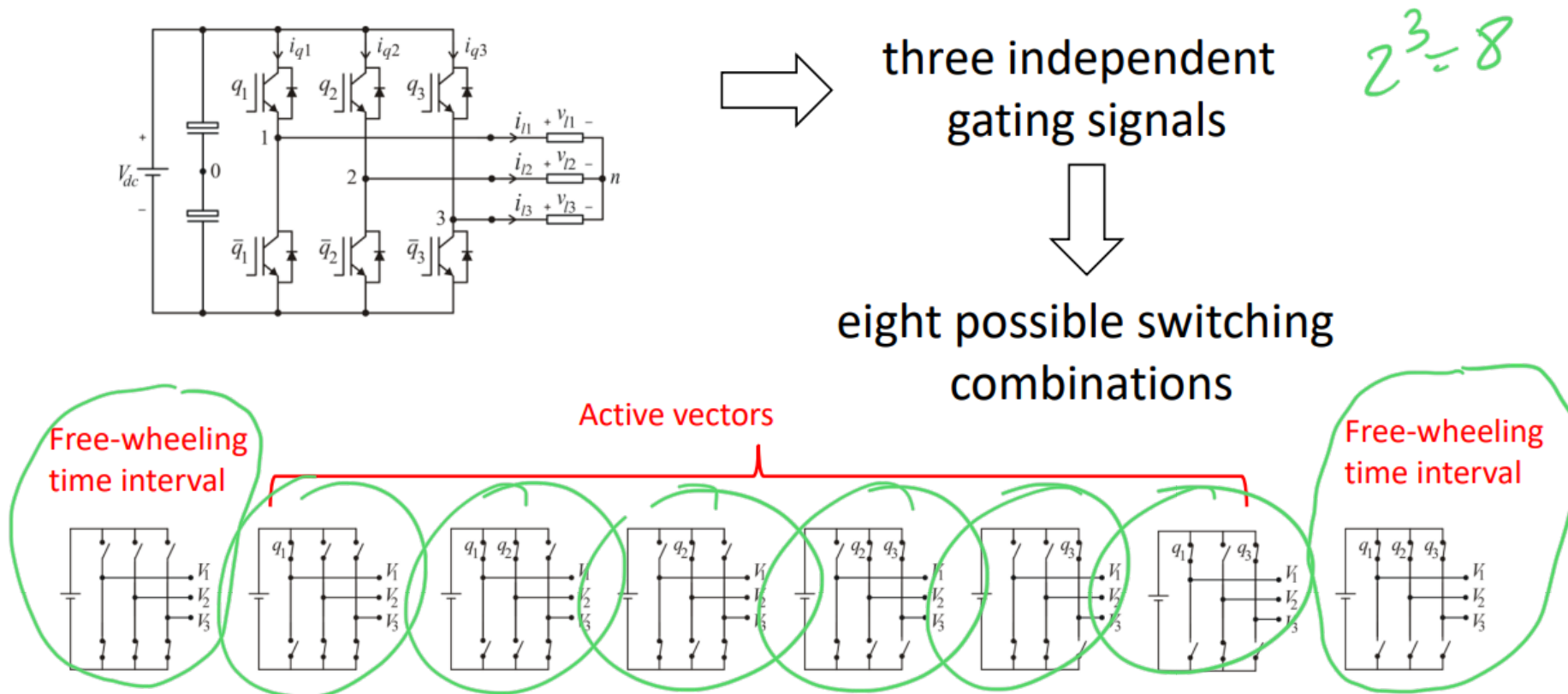


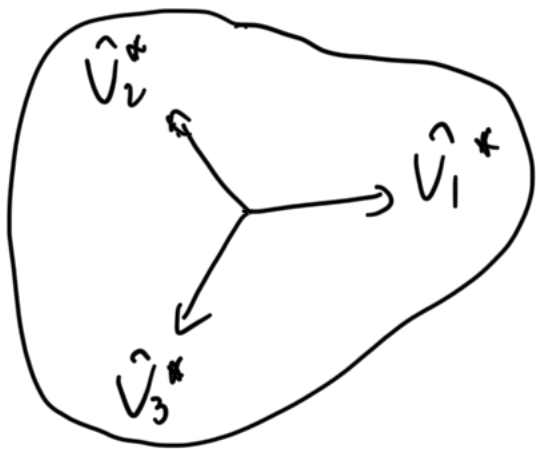
SINE PWM (SPWM)



Space Vector Modulation

The non-sinusoidal carrier-based PWM (CPWM) techniques presented in this chapter could be implemented by either an analog or a digital way. On the other hand, the space vector modulation is an inherent digital way to generate the gating signals





Space Vector Modulation

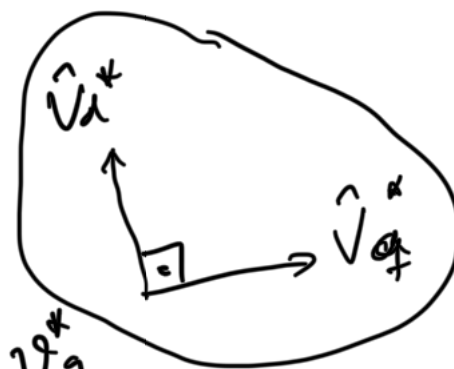
$$\bar{v}_{odq}^s = \bar{P} \bar{v}_{123}^s$$

$$v_d^s = \sqrt{\frac{2}{3}} \left(v_1^s - \frac{1}{2} v_2^s - \frac{1}{2} v_3^s \right)$$

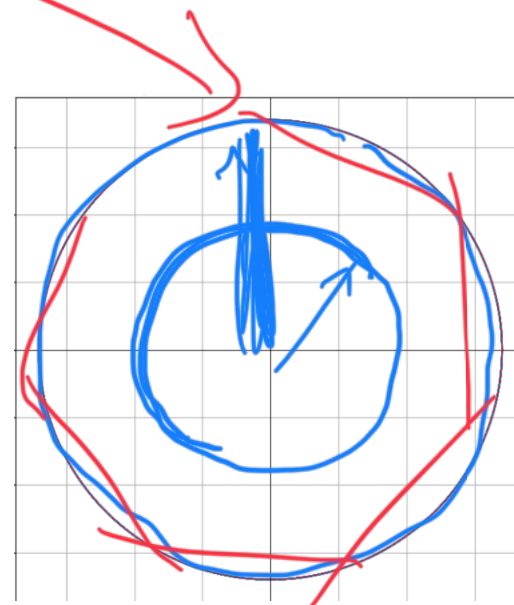
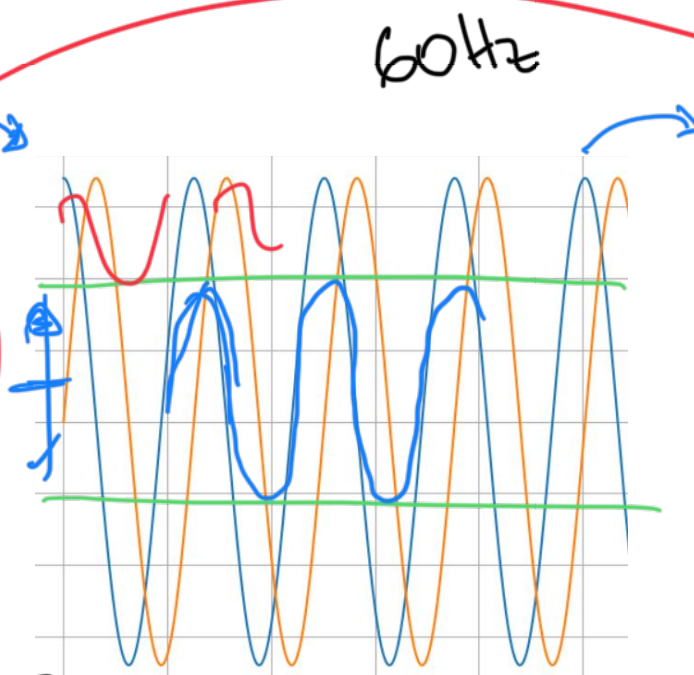
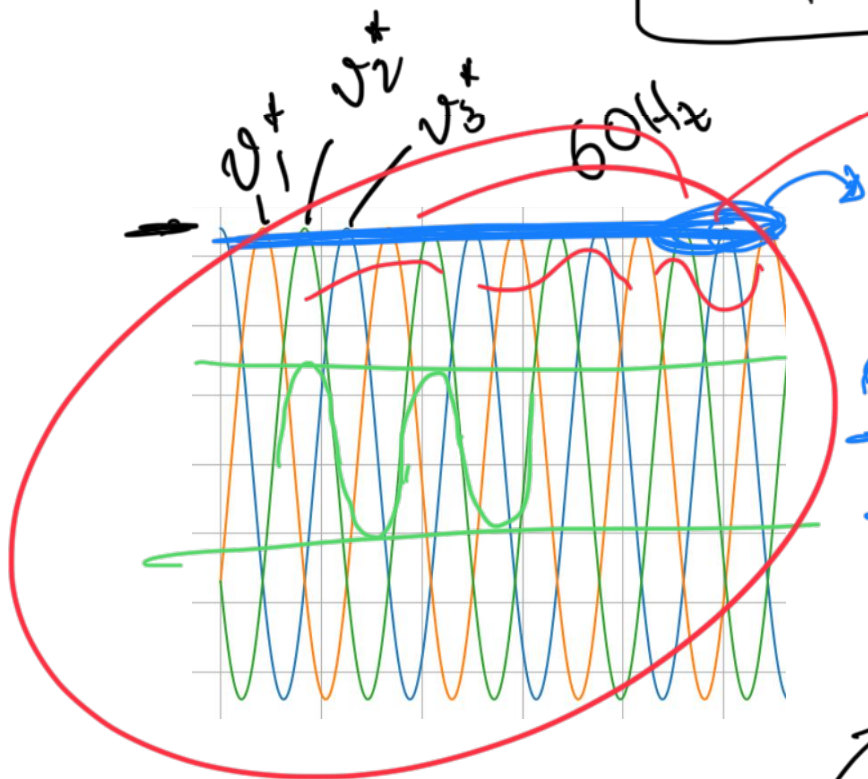
$$v_q^s = \sqrt{\frac{2}{3}} \left(\frac{\sqrt{3}}{2} v_2^s - \frac{\sqrt{3}}{2} v_3^s \right)$$

$$v_1^s + v_2^s + v_3^s = 0$$

v_0^s is null



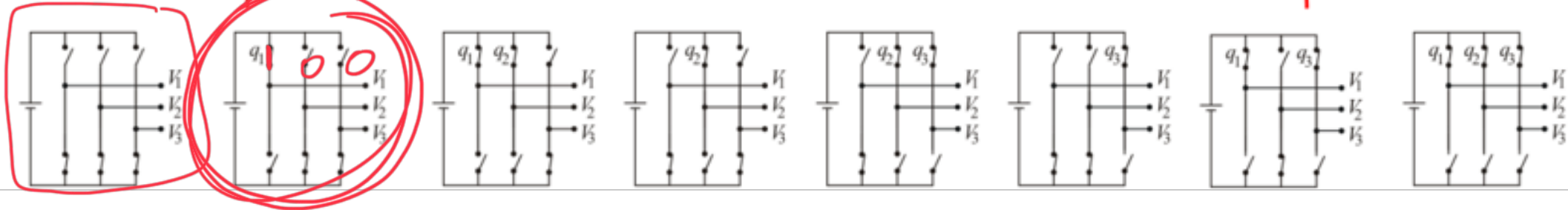
$$v_{dq}^* = v_d^* + j v_q^*$$



Free-wheeling
time interval

Active vectors

Free-wheeling
time interval



Each switching state will define a vector in the dq plane

$$v_d^s = \sqrt{\frac{2}{3}}(q_1 - \frac{1}{2}q_2 - \frac{1}{2}q_3)V_{dc}$$

$$v_q^s = \frac{1}{\sqrt{2}}(q_2 - q_3)V_{dc}$$

$$\begin{matrix} q_1 & q_2 & q_3 \\ 1 & 0 & 0 \end{matrix}$$

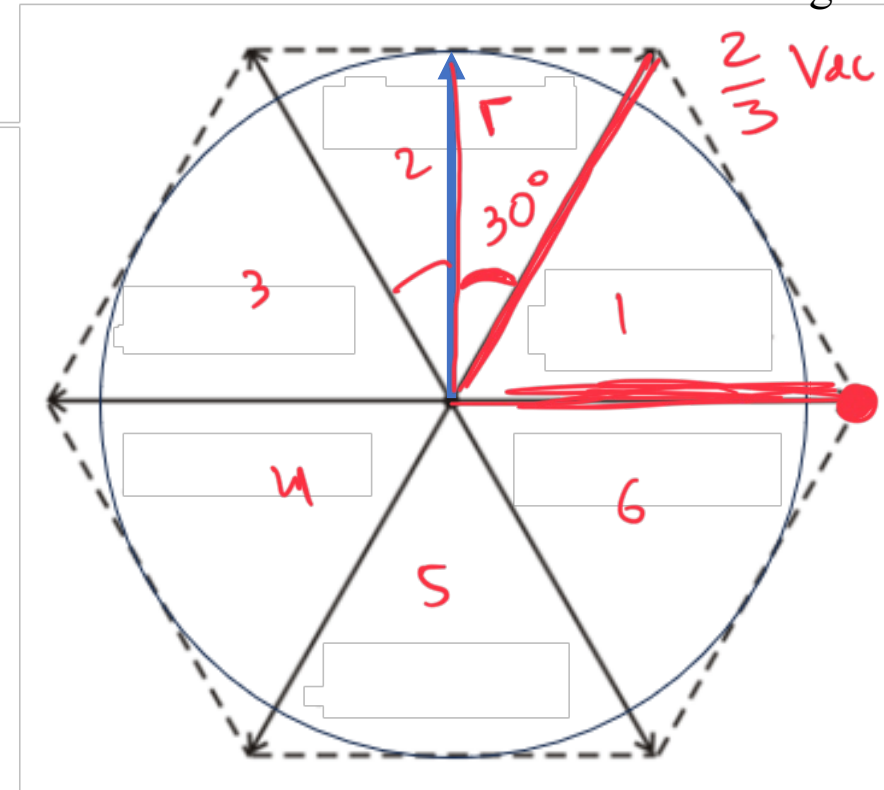
$$V_{10} = V_1 + V_{10}$$

$$V_1 = V_{10} - \underline{V_{10}}$$

$$V_1 = \frac{2}{3}V_{10} - \frac{1}{3}V_{20} - \frac{1}{3}V_{30}$$

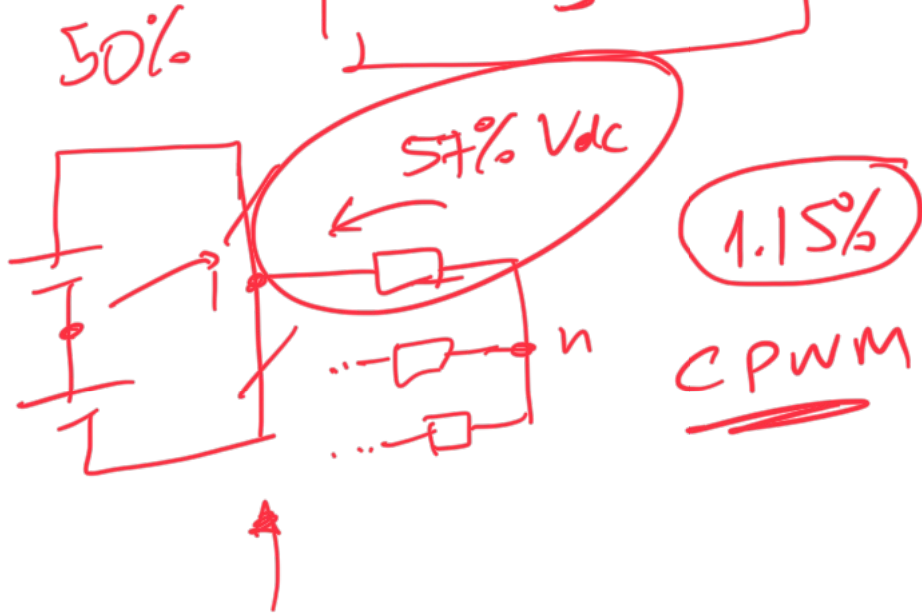
$$V_1 = \frac{2}{3} \cdot \frac{V_{dc}}{2} - \frac{1}{3}(-\frac{V_{dc}}{2}) - \frac{1}{3}(-\frac{V_{dc}}{2})$$

Maximum voltage v_{l1} (15% +)



$$v_1 = \frac{V_{dc}}{3} + \frac{V_{dc}}{3}$$

$$v_1 = \frac{2}{3} V_{dc}$$



$$\underline{0.66 V_{dc}}$$

$$\Gamma = \frac{2}{3} V_{dc} \cdot \cos(30^\circ)$$

$$\Gamma = \frac{2}{3} V_{dc} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{3} V_{dc} = \frac{1}{\sqrt{3}} V_{dc}$$

$$\Gamma = \frac{V_{dc}}{\sqrt{3}} = \underline{0.577 V_{dc}}$$

Space Vector Modulation

From the electrical machine and reference-frame theory analyses:
it is possible to define a transformation to represent the original three-phase variables in a simple system with variables in quadrature.

$$\bar{v}_{odq}^s = \bar{P} \bar{v}_{123}^s$$

stationary reference-frame

Such a transformation changes the variables with no physical connotation.

$$\bar{v}_{odq}^s = \begin{bmatrix} \bar{v}_o^s \\ \bar{v}_d^s \\ \bar{v}_q^s \end{bmatrix}, \bar{v}_{123}^s = \begin{bmatrix} \bar{v}_1^s \\ \bar{v}_2^s \\ \bar{v}_3^s \end{bmatrix}, \text{ and } \bar{P} = \sqrt{\frac{2}{3}} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}.$$

Space Vector Modulation

$$\bar{v}_{odq}^s = \bar{P} \bar{v}_{123}^s$$

$$\left| \begin{array}{l} v_1^s + v_2^s + v_3^s = 0 \\ v_o^s \text{ is null} \end{array} \right.$$

$$v_d^s = \sqrt{\frac{2}{3}} \left(v_1^s - \frac{1}{2} v_2^s - \frac{1}{2} v_3^s \right)$$

$$v_q^s = \sqrt{\frac{2}{3}} \left(\frac{\sqrt{3}}{2} v_2^s - \frac{\sqrt{3}}{2} v_3^s \right)$$

Each switching state will define a vector in the dq plane

$$v_d^s = \sqrt{\frac{2}{3}} \left(q_1 - \frac{1}{2} q_2 - \frac{1}{2} q_3 \right) V_{dc}$$

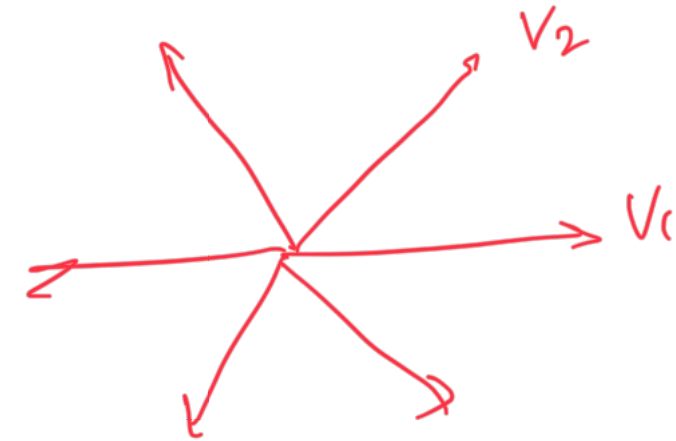
$$v_q^s = \frac{1}{\sqrt{2}} (q_2 - q_3) V_{dc}$$

Space Vector Modulation

Table 3 shows the vectors in the dq plane $V_x^s = V_d^s + jV_q^s$ ($x = 1, 2 \dots 8$) associated with each switching state. Each vector means a topological circuit for the converter.

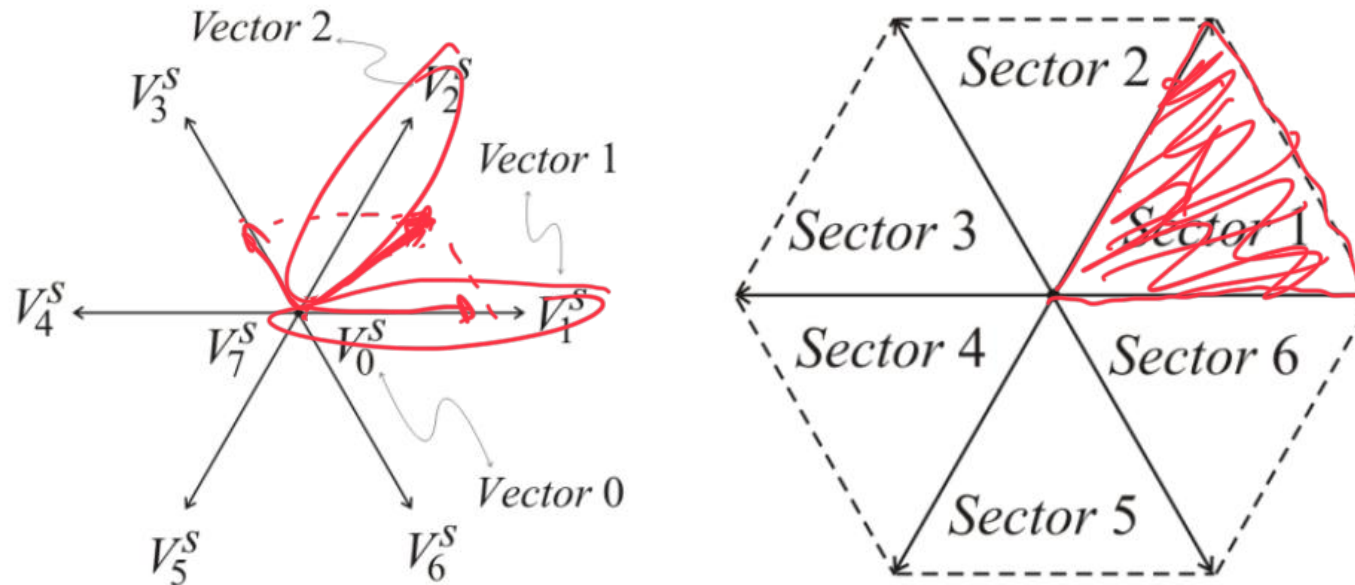
Table 3. PWM vectors as a function of the switching states.

Vector	q_1	q_2	q_3	V_x^s
0	0	0	0	0
1	1	0	0	$\sqrt{2/3}V_{dc}$
2	1	1	0	$V_{dc}/\sqrt{6} + jV_{dc}/\sqrt{2}$
3	0	1	0	$-V_{dc}/\sqrt{6} + jV_{dc}/\sqrt{2}$
4	0	1	1	$-\sqrt{2/3}V_{dc}$
5	0	0	1	$-V_{dc}/\sqrt{6} - jV_{dc}/\sqrt{2}$
6	1	0	1	$V_{dc}/\sqrt{6} - jV_{dc}/\sqrt{2}$
7	1	1	1	0

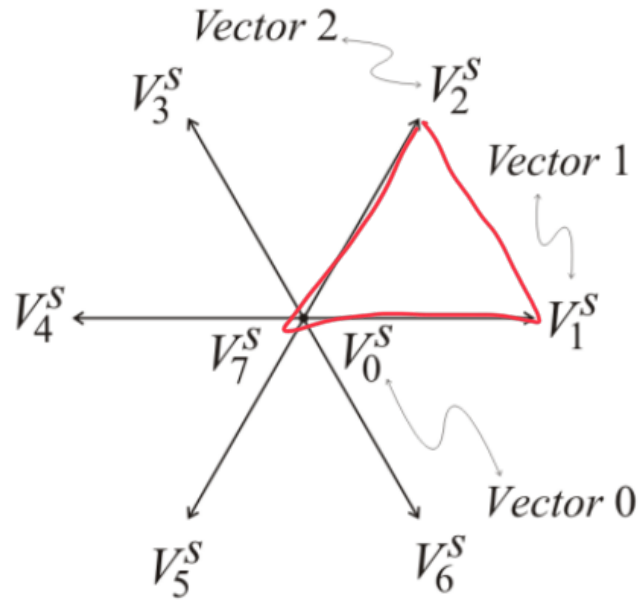


Space Vector Modulation

Notice from Table 3 that there are two null (0 and 7) and six non-null (1, 2, 3, 4, 5 and 6) vectors, which will be used to synthesize the desired load voltage. All vectors V_x^s ($x = 1, 2 \dots 8$) are presented in Fig. 24(a). The area between two adjacent non-null vectors defines a sector as considered in a graphical analysis presented in Fig. 24(b).



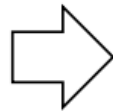
Space Vector Modulation



Interest in adjacent vectors

(2V.) (Act) (Act) (2V)

Just a single change
for the switching state



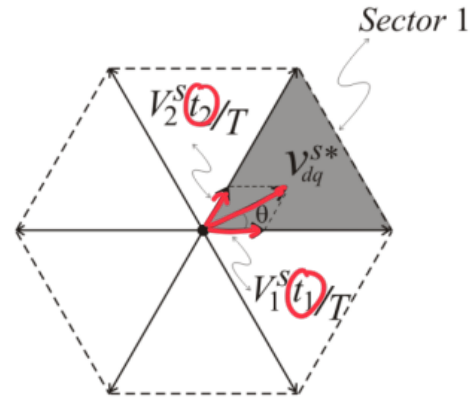
$x = 1, 2 \dots 8$

Table 3. PWM vectors as a function of the switching states.

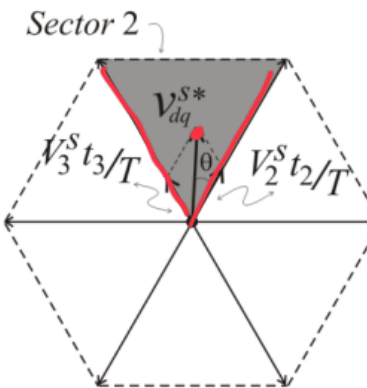
Vector	q_1	q_2	q_3	V_x^s
0	0	0	0	0
1	1	0	0	$\sqrt{2/3}V_{dc}$
2	1	1	0	$V_{dc}/\sqrt{6} + jV_{dc}/\sqrt{2}$
3	0	1	0	$-V_{dc}/\sqrt{6} + jV_{dc}/\sqrt{2}$
4	0	1	1	$-\sqrt{2/3}V_{dc}$
5	0	0	1	$-V_{dc}/\sqrt{6} - jV_{dc}/\sqrt{2}$
6	1	0	1	$V_{dc}/\sqrt{6} - jV_{dc}/\sqrt{2}$
7	1	1	1	0

Space Vector Modulation

To reduce the switching losses, if the desired voltage is inside of Sector 1, the Vectors 1 and 2 should be chosen



If the desired voltage is inside of Sector 2, the Vectors 2 and 3 should be chosen



Space Vector Modulation

The adjacent vectors can be defined in a generic way by using

$$V_k^s \text{ and } V_l^s$$

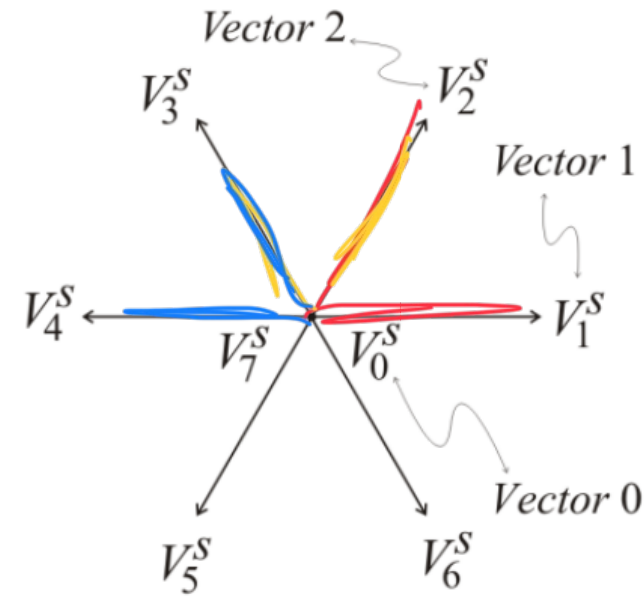
($k = 1, \dots, 6$, with $l = k + 1$ if $k \leq 5$ and $l = 1$ if $k = 6$)

The vectors V_k^s and V_l^s have their dq components

$$\begin{aligned} V_k^s &= V_{dk}^s + jV_{qk}^s \\ V_l^s &= V_{dl}^s + jV_{ql}^s \end{aligned}$$

Table 3. PWM vectors as a function of the switching states.

Vector	q_1	q_2	q_3	V_x^s
0	0	0	0	0
1	1	0	0	$\sqrt{2/3}V_{dc}$
2	1	1	0	$V_{dc}/\sqrt{6} + jV_{dc}/\sqrt{2}$
3	0	1	0	$-V_{dc}/\sqrt{6} + jV_{dc}/\sqrt{2}$
4	0	1	1	$-\sqrt{2/3}V_{dc}$
5	0	0	1	$-V_{dc}/\sqrt{6} - jV_{dc}/\sqrt{2}$
6	1	0	1	$V_{dc}/\sqrt{6} - jV_{dc}/\sqrt{2}$
7	1	1	1	0



$x = 1, 2 \dots 8$

Space Vector Modulation

The voltage v_{dq}^{s*} is obtained from a set of three-phase reference voltages (v_{123}^{s*}) by using the transformation $\bar{P}$ given

$$\bar{v}_{odq}^s = \begin{bmatrix} \bar{v}_o^s \\ \bar{v}_d^s \\ \bar{v}_q^s \end{bmatrix}, \bar{v}_{123}^s = \begin{bmatrix} \bar{v}_1^s \\ \bar{v}_2^s \\ \bar{v}_3^s \end{bmatrix}, \text{ and } \bar{P} = \sqrt{\frac{2}{3}} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}.$$

a switching sequence for q_1 , q_2 and q_3 should be chosen to allow the converter to synthesize a given reference voltage v_{dq}^{s*} , which is a constant in the switching period (T).

$$\frac{1}{T} \int_0^T v_{dq}^{s*} dt = \left[\frac{1}{T} \int_0^{t_k} V_k^s dt + \frac{1}{T} \int_0^{t_l} V_l^s dt \right]$$

$$v_{dq}^{s*} = \left[\frac{t_k}{T} V_k^s + \frac{t_l}{T} V_l^s \right]$$

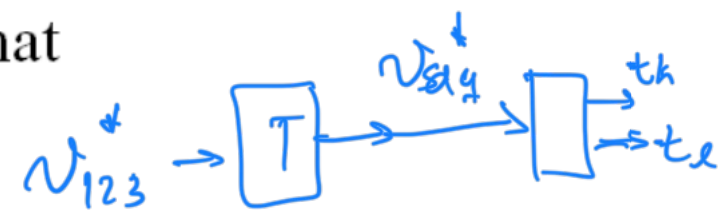
Space Vector Modulation

$$\left[\begin{aligned} \frac{1}{T} \int_0^T v_{dq}^{s*} dt &= \frac{1}{T} \int_0^{t_k} V_k^s dt + \frac{1}{T} \int_0^{t_l} V_l^s dt \\ v_{dq}^{s*} &= \frac{t_k}{T} V_k^s + \frac{t_l}{T} V_l^s \end{aligned} \right.$$

where t_k and t_l are interval of time for application of the vectors V_k^s and V_l^s .

Rewriting in terms of dq components, it turns out that

$$\left[\begin{aligned} v_d^{s*} &= \frac{t_k}{T} V_{kd}^s + \frac{t_l}{T} V_{ld}^s \\ v_q^{s*} &= \frac{t_k}{T} V_{kq}^s + \frac{t_l}{T} V_{lq}^s \end{aligned} \right. \Rightarrow \left\{ \begin{aligned} t_k &= \frac{v_d^{s*} V_{lq}^s - v_q^{s*} V_{ld}^s}{V_{kd}^s V_{lq}^s - V_{ld}^s V_{kq}^s} T \\ t_l &= \frac{v_q^{s*} V_{kd}^s - v_d^{s*} V_{kq}^s}{V_{kd}^s V_{lq}^s - V_{ld}^s V_{kq}^s} T \end{aligned} \right.$$



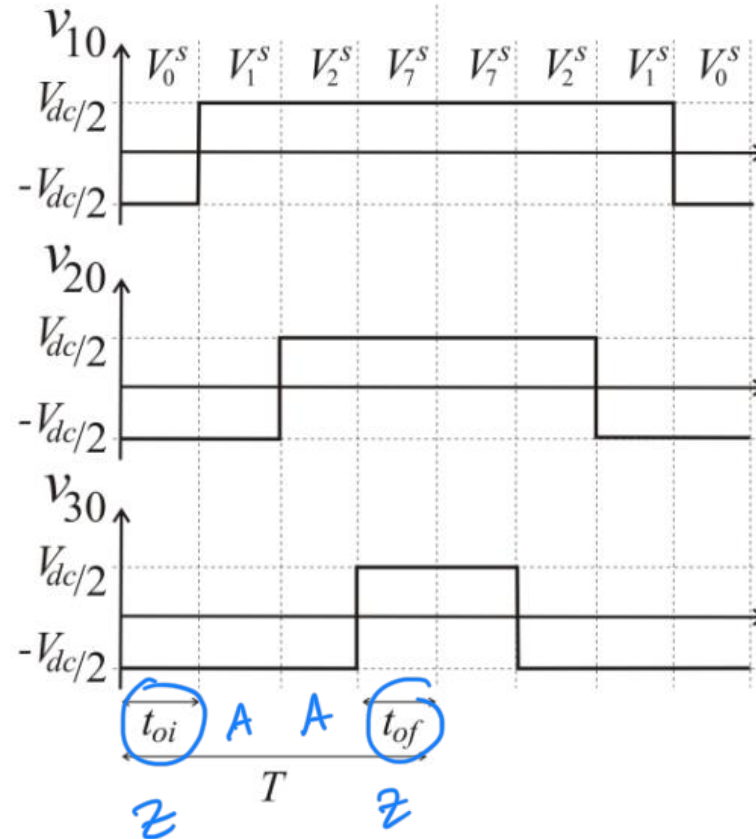
To keep the switching frequency of the converter constant, it is necessary to guarantee that the sum of the time for the application of the vectors is equal to T . Then, the null vectors (Vectors 0 and 7 in Table 3) can be used to keep the switching frequency constant, i.e., $t_k + t_l + t_o = T$, and then $t_o = T - t_k - t_l$.

Space Vector Modulation

Notice that there is a degree of freedom for t_o ,
 this time could be divided between the beginning ($t_{oi} = \mu t_o$)
 and the end [$t_{of} = (1 - \mu)t_o$, $0 \leq \mu \leq 1$]
 of the sampling interval.

$$t_k = \frac{v_d^{s*} V_{lq}^s - v_q^{s*} V_{ld}^s}{V_{kd}^s V_{lq}^s - V_{ld}^s V_{qk}^s} T$$

$$t_l = \frac{v_q^{s*} V_{kd}^s - v_d^{s*} V_{kq}^s}{V_{kd}^s V_{lq}^s - V_{ld}^s V_{qk}^s} T$$



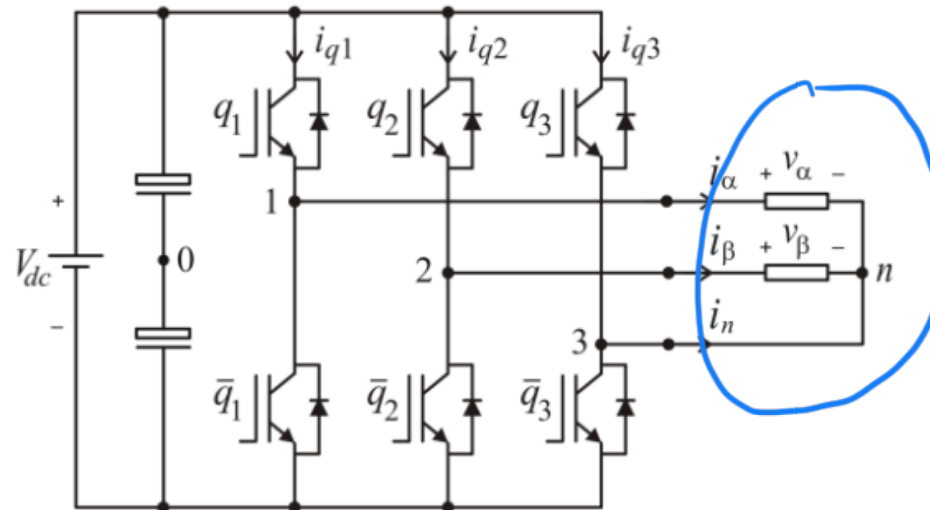
5. Other Configurations with Hybrid PWM

5.1. Three-leg Converter – Two-phase machine

$$\begin{aligned}v_{10}^* &= v_{\alpha}^* + v_h^* \\v_{20}^* &= v_{\beta}^* + v_h^* \\v_{30}^* &= v_h^*\end{aligned}$$

$$v_h^* = V_{dc} \left(\mu - \frac{1}{2} \right) + (\mu - 1)V_{max}^* - \mu V_{min}^*$$

$$\begin{aligned}V_{max} &= \text{MAX}\{v_{\alpha}^*, v_{\beta}^*, 0\} \\V_{min} &= \text{MIN}\{v_{\alpha}^*, v_{\beta}^*, 0\}\end{aligned}$$



$$v_h^k = f(\mu, v_{max}, v_{min}, v_{dc})$$

CPM W

$$v_{10}^k = v_1^k + v_h^k$$

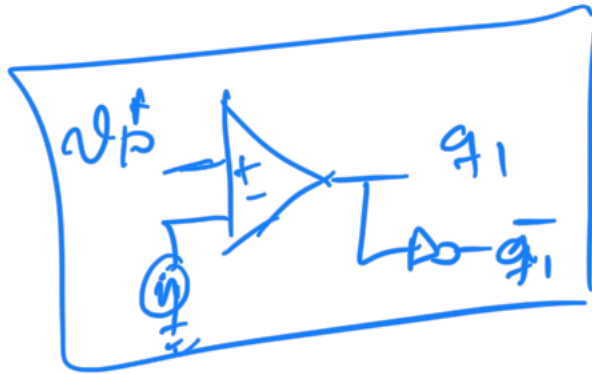
$$v_{20}^k = v_2^k + v_h^k$$

⋮

$$v_{60}^k = v_6^k + v_h^k$$

$$v_{max} = \max \{ v_1^k, v_2^k, \dots, v_6^k \}$$

$$v_{min} = \min \{ \quad \quad \quad \}$$



$$\sigma_1 = \left(\frac{v_{10}^k}{v_{dc}} + \frac{1}{2} \right) \cdot \tau_s$$

